

§7-5 三相电路中的功率

➤ 不对称三相电路

$$P = P_A + P_B + P_C$$

$$Q = Q_A + Q_B + Q_C$$

$$S = \sqrt{P^2 + Q^2}$$

$$\tilde{S} = \tilde{S}_A + \tilde{S}_B + \tilde{S}_C$$

例1. 一台三相电动机的额定相电压为380 V，作三角形联结时，额定线电流为19 A，额定功率为10 kW。(1) 求这台电动机的功率因数及每相阻抗，并求其在额定工作状态下所吸收的无功功率；(2) 如果把这台电动机改接成星形(假定每相阻抗不变)，仍然联接到线电压为380 V的对称三相电源上，则线电流及电动机吸收的功率将如何改变？

解：(1)

$$\cos \varphi = \frac{P}{\sqrt{3} U_l I_l} = \frac{10 \times 10^3}{\sqrt{3} \times 380 \times 19} = 0.8$$

$$\varphi = \arccos 0.8 = 36.9^\circ \quad \text{阻抗角}$$

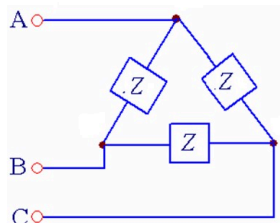
$$I_p = \frac{I_l}{\sqrt{3}} = \frac{19}{\sqrt{3}} \text{ A} = 11 \text{ A}$$

$$|Z| = \frac{U_p}{I_p} = \frac{380}{11} \Omega = 34.6 \Omega$$

$$Z = |Z| \angle \varphi = 34.6 \angle 36.9^\circ \Omega$$

$$U_l = U_p = 380 \text{ V}$$

$$Q = \sqrt{3} U_l I_l \sin \varphi = \sqrt{3} \times 380 \times 19 \times \sin 36.9^\circ = 7503 \text{ var} = 7.503 \text{ k var}$$



➤ 对称三相电路

$$P = 3P_p = 3U_p I_p \cos \varphi_p = \sqrt{3} U_l I_l \cos \varphi_p$$

$$\text{对于Y形联结对称三相电路} \quad U_p = \frac{1}{\sqrt{3}} U_l \quad I_p = I_l$$

$$\text{对于}\Delta\text{形联结对称三相电路} \quad U_p = U_l \quad I_p = \frac{1}{\sqrt{3}} I_l$$

$$Q = 3U_p I_p \sin \varphi_p = \sqrt{3} U_l I_l \sin \varphi_p$$

$$S = 3U_p I_p = \sqrt{3} U_l I_l$$

注意：

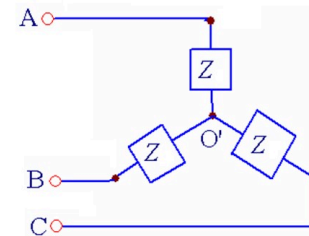
无论是用相电压和相电流计算功率还是用线电压和线电流计算功率，

φ_p 均指某一相电压超前于同一相电流的相角。

φ_p 也是每相负载的阻抗角。

$$(2) \quad U'_p = \frac{U_l}{\sqrt{3}} = \frac{380}{\sqrt{3}} \text{ V} = 220 \text{ V}$$

$$I'_l = I'_p = \frac{U'_p}{|Z|} = \frac{220}{34.6} \text{ A} = 6.36 \text{ A}$$



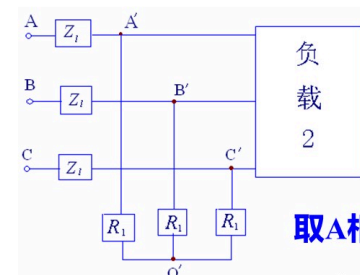
作星形联结时的线电流为作三角形联结时线电流的三分之一

电动机吸收的功率

$$P_Y = \sqrt{3} U'_l I'_l \cos \varphi = \sqrt{3} \times 380 \times 6.36 \times 0.8 = 3340 \text{ W} = 3.34 \text{ kW}$$

作星形联结时的功率等于作三角形联结时功率的三分之一

例2. 下图所示对称三相电路中, 已知第一组负载 $R_1=110\Omega$, 额定电压为380/220V; 第二组负载为**对称感性负载**, 其功率 $P=5.28\text{kW}$, 功率因数为0.8, 额定电压也是380/220V。线路阻抗 $Z_l=1+j2\Omega$ 。求这两组负载额定运行时, 电源端的线电压的大小。



解: 对于第二组负载

$$I_{2l} = \frac{P}{\sqrt{3}U_{2l}\cos\varphi_2} = \frac{5280}{\sqrt{3} \times 380 \times 0.8} = 10\text{A}$$

对称感性负载 $\varphi_2 = \arccos 0.8 = 36.9^\circ$

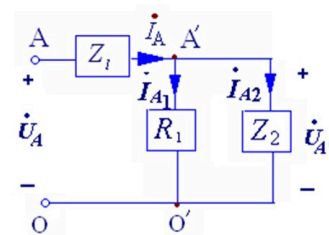
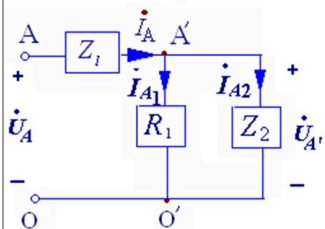
取A相计算, 令 $\dot{U}_{A'} = 220\angle 0^\circ\text{V}$ $\dot{U}_{A'B} = 380\angle 30^\circ\text{V}$

$\dot{I}_{A2} = 10\angle(0^\circ - 36.9^\circ)\text{A}$ **负载星形联接**

$\dot{I}_{A2} = 10\angle(30^\circ - 36.9^\circ - 30^\circ)\text{A}$ **负载三角形联接**

$$\dot{I}_{A1} = \frac{220\angle 0^\circ}{110} = 2\angle 0^\circ\text{A}$$

$$\dot{I}_A = \dot{I}_{A1} + \dot{I}_{A2} = 2 + 10\angle -36.9^\circ = 10 - j6\text{A}$$



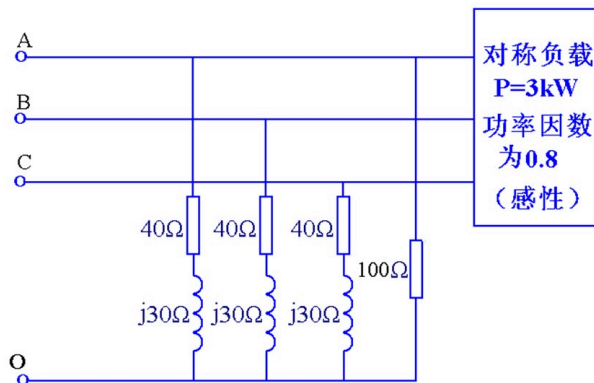
$$\begin{aligned}\dot{U}_A &= Z_l \dot{I}_A + \dot{U}_{A'} \\ &= (1 + j2)(10 - j6) + 220 \\ &= 242.4\angle 3.3^\circ\text{V}\end{aligned}$$

故:

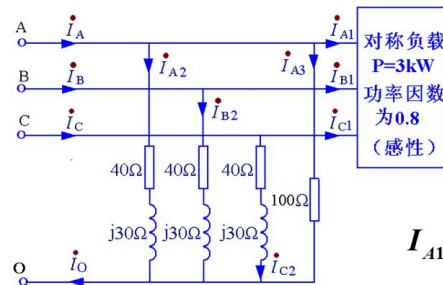
$$U_l = \sqrt{3}U_A = \sqrt{3} \times 242.4 = 420\text{V}$$

例3. 图示380V/220V的三相四线制供电系统中接有两个对称三相负载和一个单相负载, 试问

(1) 三个线电流各为多少? (2) 中线上有无电流, 若有, 应为多少?



解: 设定各相关电流如下图所示



可将负载考虑为三角形联接

设 $\dot{U}_{AB} = 380\angle 0^\circ$

$\varphi = \arccos 0.8 = 36.9^\circ$ (感性)

$$I_{A1} = \frac{P}{\sqrt{3}U_{AB}\cos\varphi} = 5.7\text{A}$$

$$\dot{I}_{A1} = I_{A1}\angle -\varphi - 30^\circ = 5.7\angle -66.9^\circ\text{A}$$

$$\dot{I}_{B1} = \dot{I}_{A1}\angle -120^\circ = 5.7\angle 173.1^\circ\text{A}$$

$$\dot{I}_{C1} = \dot{I}_{A1}\angle 120^\circ = 5.7\angle 53.1^\circ\text{A}$$

对称负载
P=3kW
功率因数
为0.8
(感性)

$$\dot{U}_{AO} = \frac{\dot{U}_{AB}}{\sqrt{3}} \angle -30^\circ \text{ V}$$

$$\dot{I}_{A2} = \frac{\dot{U}_{AO}}{40 + j30} = 4.4 \angle -66.9^\circ \text{ A}$$

$$\dot{I}_{B2} = \dot{I}_{A2} \angle -120^\circ = 4.4 \angle 173.1^\circ \text{ A}$$

$$\dot{I}_{C2} = \dot{I}_{A2} \angle 120^\circ = 4.4 \angle 53.1^\circ \text{ A}$$

$$\dot{I}_{A3} = \frac{\dot{U}_{AO}}{100} = 2.2 \angle -30^\circ \text{ A}$$

$$\dot{I}_A = \dot{I}_{A1} + \dot{I}_{A2} + \dot{I}_{A3} = 11.93 \angle -60.6^\circ \text{ A}$$

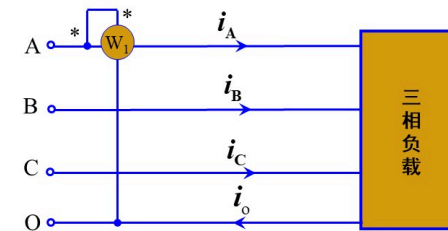
$$\dot{I}_B = \dot{I}_{B1} + \dot{I}_{B2} = 10.1 \angle 173.1^\circ \text{ A}$$

$$\dot{I}_C = \dot{I}_{C1} + \dot{I}_{C2} = 10.1 \angle 53.1^\circ \text{ A}$$

$$\dot{I}_O = \dot{I}_{A3} = 2.2 \angle -30^\circ \text{ A}$$

三相功率的测量

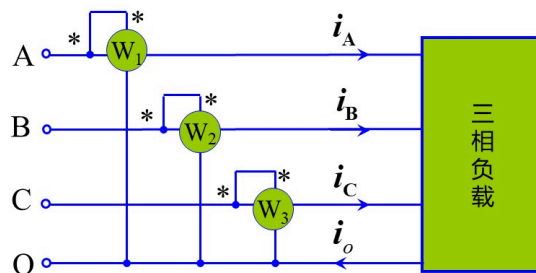
◆对于三相四线制电路，负载对称用一瓦计法 ($P=3P_A$)



功率表的读数 $P_A = U_A I_A \cos \varphi = \text{Re}[\dot{U}_A \cdot \dot{I}_A^*]$

三相功率的测量

◆对于三相四线制电路，不对称用三瓦计法

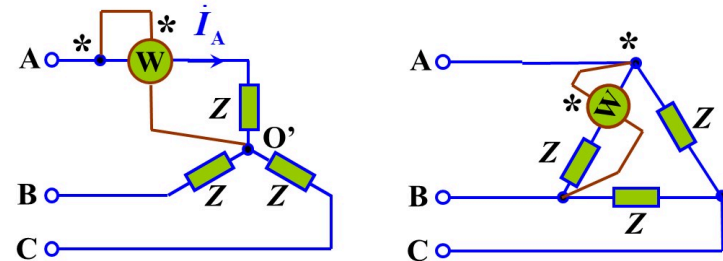


$$p = u_A i_A + u_B i_B + u_C i_C$$

$$P = P_1 + P_2 + P_3$$

三相功率的测量

◆对于三相三线制电路，负载对称用一瓦计法 ($P=3P_A$)

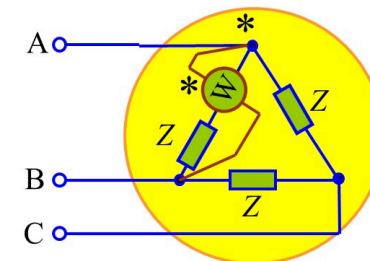
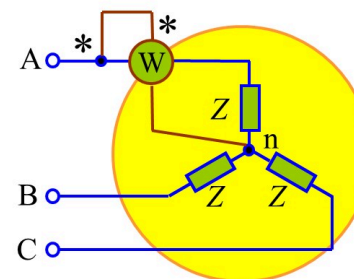


一瓦计法的适用范围：对称三相电路（三线制/四线制皆可）

下列电路中适合采用“一表法”测量的是

- ☐ A 对称三相三线制电路
- ☐ B 不对称三相三线制电路
- ☐ C 对称三相四线制电路
- ☐ D 不对称三相四线制电路

三相功率的测量

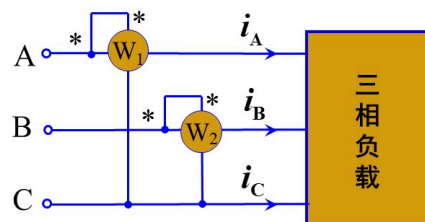


➤ 中性点在外壳内部

➤ 绕组不宜拆开串接功率表



◆对于三相三线制，不管对称与否均用二瓦计法



$$P = P_1 + P_2$$

一个表的读数无意义

$$p_A = u_A i_A$$

$$p_B = u_B i_B$$

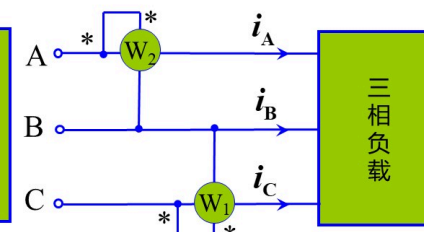
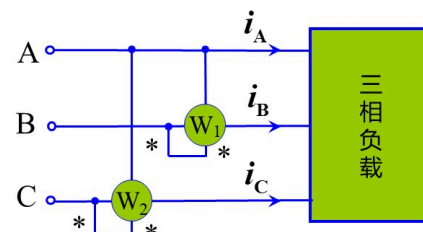
$$p_C = u_C i_C = u_C (-i_A - i_B) = u_{AC} i_A + u_{BC} i_B$$

$$p = p_A + p_B + p_C$$

$$= u_A i_A + u_B i_B + u_C (-i_A - i_B)$$

$$\text{条件: } i_A + i_B + i_C = 0$$

◆二瓦计法不同接法



若 W_1 的读数为 P_1 ， W_2 的读数为 P_2 ，
则三相总功率为：

$$P = P_1 + P_2$$

例 对称三相电路, $\dot{U}_{AB} = 380\angle 30^\circ \text{V}$, $\dot{I}_A = 7.56\angle -46.2^\circ \text{A}$
求: 两表读数及对称三相负载吸收的总功率。

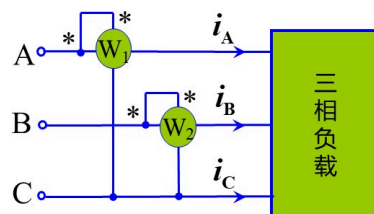
解: $\dot{U}_{AC} = -\dot{U}_{CA}$
 $= -380\angle (30^\circ + 120^\circ) \text{V}$
 $= 380\angle (-30^\circ) \text{V}$
 $\dot{U}_{BC} = 380\angle (30^\circ - 120^\circ) \text{V}$
 $= 380\angle (-90^\circ) \text{V}$

$$\dot{I}_B = 7.56\angle (-46.2^\circ - 120^\circ) \text{A} = 7.56\angle (-166.2^\circ) \text{A}$$

$$P_1 = \text{Re}[\dot{U}_{AC} \cdot \dot{I}_A^*] = 380 \times 7.56 \times \cos(-30^\circ + 46.2^\circ) \text{W} = 2758.73 \text{W}$$

$$P_2 = \text{Re}[\dot{U}_{BC} \cdot \dot{I}_B^*] = 380 \times 7.56 \times \cos(-90^\circ + 166.2^\circ) \text{W} = 685.27 \text{W}$$

$$P = P_1 + P_2 = (2758.73 + 685.27) \text{W} = 3444 \text{W}$$



投票 最多可选1项

若图中两个功率表的读数分别为1500W和 - 600W, 则该对称三相电路的三相电源发出的总功率为



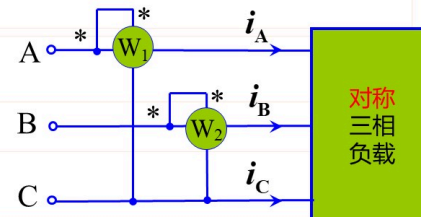
900W



2100W



题目有错吧, 怎么有负值?

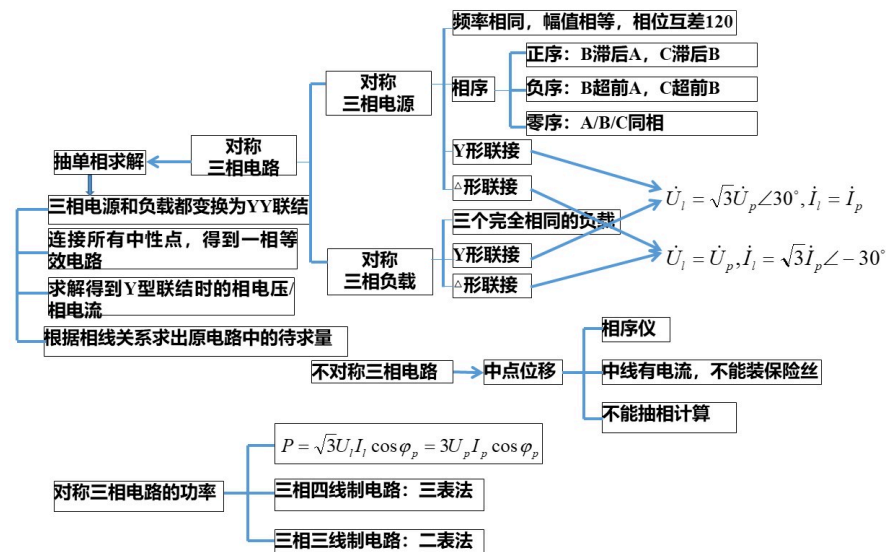


➤ 对称三相电路瞬时功率的平衡性

$$\begin{aligned} p(t) &= p_A(t) + p_B(t) + p_C(t) \\ &= \sqrt{2}U \sin \omega t \cdot \sqrt{2}I \sin(\omega t - \varphi) \\ &\quad + \sqrt{2}U \sin(\omega t - \frac{2}{3}\pi) \cdot \sqrt{2}I \sin(\omega t - \varphi - \frac{2}{3}\pi) \\ &\quad + \sqrt{2}U \sin(\omega t + \frac{2}{3}\pi) \cdot \sqrt{2}I \sin(\omega t - \varphi + \frac{2}{3}\pi) \\ &= UI[\cos \varphi - \cos(2\omega t - \varphi)] \\ &\quad + UI[\cos \varphi - \cos(2\omega t - \varphi - \frac{4}{3}\pi)] \\ &\quad + UI[\cos \varphi - \cos(2\omega t - \varphi + \frac{4}{3}\pi)] \\ &= 3UI \cos \varphi = P \end{aligned}$$

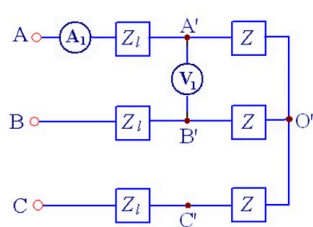
对称三相制的瞬时功率的平衡性是对称三相制的重要优点之一, 它决定了三相旋转电机在对称情形下运行时其**瞬时转矩恒定的特性**。

三相电路知识点汇总图



课堂练习:

1、图示对称Y-Y联接三相电路中，电压表的读数为380V，电流表的读数为10A， $Z_l = (3+j4)\Omega$ ，三相感性负载Z吸收的有功功率为6000W，求Z及电源线电压 U_{AB} 。



解：负载的功率因数

$$\cos \varphi = \frac{P}{\sqrt{3} U_{A'B'} I_A} = \frac{6000}{\sqrt{3} \times 380 \times 10} = 0.912$$

$$\varphi = 24.27^\circ \quad (\text{感性})$$

$$|Z| = \frac{U_{A'}}{I_A} = \frac{220}{10} = 22\Omega$$

$$Z = |Z| \angle \varphi = 22 \angle 24.27^\circ = 20 + j9(\Omega)$$

三相化作单相计算

令

$$\dot{U}_{A'O'} = 220 \angle 0^\circ \text{ V}$$

则

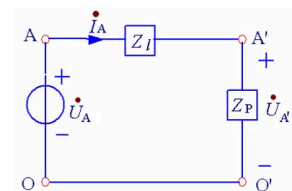
$$\dot{I}_A = 10 \angle -24.27^\circ \text{ A}$$

$$\begin{aligned} \dot{U}_A &= \dot{I}_A (Z_l + Z) = 10 \angle -24.27^\circ \times (3 + j4 + 20 + j9) \\ &= 10 \angle -24.27^\circ \times 26.42 \angle 29.48^\circ = 264.2 \angle 5.21^\circ \text{ V} \end{aligned}$$

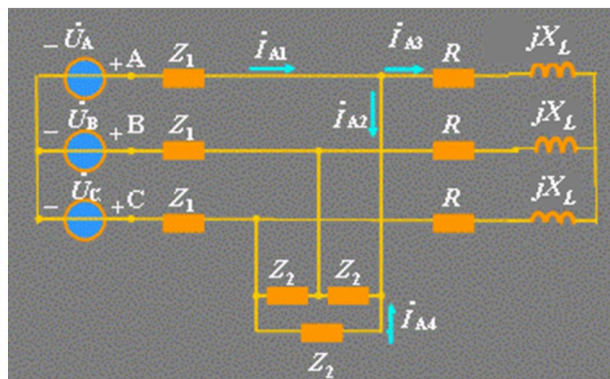
$$U_{AB} = \sqrt{3} U_A = \sqrt{3} \times 264.2 = 457.6 \text{ V}$$

或
$$\begin{aligned} \dot{U}_A &= \dot{I}_A Z_l + \dot{U}_{A'O'} = 10 \angle -24.27^\circ \times (3 + j4) + 220 \angle 0^\circ \\ &= 50 \angle 28.86^\circ + 220 \angle 0^\circ = 264.89 \angle 5.23^\circ \text{ V} \end{aligned}$$

$$U_{AB} = \sqrt{3} U_A = \sqrt{3} \times 264.89 = 458.8 \text{ V}$$



2、下图表示同时接有三角形和星形负载的对称三相电路，令 $\dot{U}_A = 220 \angle 0^\circ$ ，试求电流 $\dot{I}_{A1}, \dot{I}_{A2}, \dot{I}_{A3}, \dot{I}_{A4}$ ，电源发出的功率，以及各组负载消耗的功率。



已知：

$$Z_1 = (1 + j)\Omega$$

$$Z_2 = 120\Omega$$

$$R = 30\Omega$$

$$X_L = 40\Omega$$

解：对于线路阻抗

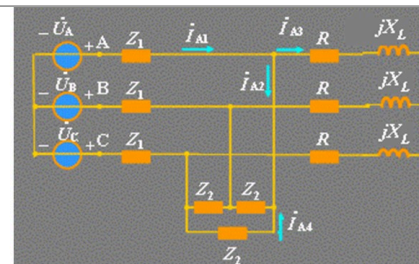
$$Z_1 = 1 + j1 = 1.41 \angle 45^\circ \Omega$$

对于第一组负载

$$Z_{2Y} = \frac{1}{3} Z_2 = 40\Omega$$

对于第二组负载

$$Z_3 = 30 + j40 = 50 \angle 53.13^\circ \Omega$$



$$\dot{I}_{A1} = \frac{\dot{U}_A}{Z_1 + Z_{2Y} \parallel Z_3} = \frac{220 \angle 0^\circ}{1.41 + j1.41 \parallel 40 \parallel 50 \angle 53.13^\circ} = 8.42 \angle -24.53^\circ \text{ A}$$

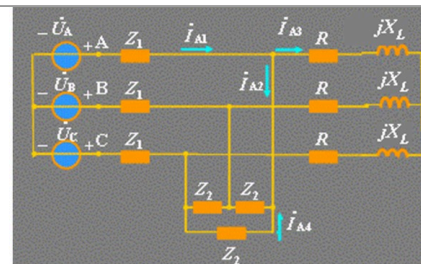
$$\begin{aligned} \dot{I}_{A2} &= \dot{I}_{A1} \frac{Z_3}{Z_{2Y} + Z_3} \\ &= 8.42 \angle -24.53^\circ \frac{50 \angle 53.13^\circ}{40 + 50 \angle 53.13^\circ} = 5.22 \angle -1.14^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} \dot{I}_{A3} &= \dot{I}_{A1} \frac{Z_{2Y}}{Z_{2Y} + Z_3} \\ &= 4.18 \angle -54.27^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} \dot{I}_{A4} &= \frac{\dot{I}_{A2} \angle 150^\circ}{\sqrt{3}} \\ &= 3.01 \angle 148.86^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} \dot{U}_{A'} &= Z_{2Y} \dot{I}_{A2} \\ &= 208.8 \angle -1.14^\circ \text{ V} \end{aligned}$$

$$\begin{aligned}\dot{I}_{A1} &= 8.42 \angle -24.53^\circ \text{ A} \\ \dot{I}_{A2} &= 5.22 \angle -1.14^\circ \text{ A} \\ \dot{I}_{A3} &= 4.18 \angle -54.27^\circ \text{ A} \\ \dot{I}_{A4} &= 3.01 \angle 148.86^\circ \text{ A} \\ \dot{U}_{A'} &= 208.8 \angle -1.14^\circ \text{ V}\end{aligned}$$



电源发出的功率为: $P_1 = 3U_{A'} I_{A1} \cos \varphi_1$
 $= 3 \times 220 \times 8.42 \times \cos 24.53^\circ = 5055.6 \text{ W}$

第一组负载吸收的功率为: $P_{21} = 3I_{A4}^2 Z_2$
 $= 3 \times 3.01^2 \times 120 = 3261.6 \text{ W}$

第二组负载吸收的功率为: $P_{22} = 3U_{A'} I_{A3} \cos \varphi_3$
 $= 3 \times 208.8 \times 4.18 \times \cos 53.13^\circ = 1571 \text{ W}$
 $(P_{22} = 3I_{A3}^2 R)$